

ENTERPRISE COMPLIANCE CONTROL PLANE

12-Hour AI Hackathon Playbook

Working product name: REGULUS

Core proposition

An AI-assisted compliance system of record and enterprise control plane that owns regulatory content, conformity, certification and evidence; verifies every required product, process and system change; and blocks market or shipment release until compliance can be proven.

Prepared for a 20-minute senior-jury demonstration

TCS AI Hackathon | Build window: 12 hours

Scenario note

The customer, systems, products, regulations and data in this playbook are synthetic hackathon assets. They may be inspired by a global powertrain manufacturer, but they do not represent any customer's actual internal architecture or confidential information.

How to use this playbook

This document is designed as a war-room manual. Use Sections 1-8 to freeze the proposition and scope, Sections 9-17 to build the product, Sections 18-22 to seed and test it, and Sections 23-27 to rehearse the jury experience.

Part	Purpose	Primary owner
A. Proposition and scope	Freeze what the team is solving and what is deliberately not being built.	Product / pitch lead
B. Functional blueprint	Define the system-of-record boundary, digital thread, snapshot, certification and governance flows.	Product + backend leads
C. Technical build	Implement Lovable experience, Supabase model, deterministic rules and adapters.	Frontend + backend leads
D. Synthetic datasets	Seed realistic enterprise-scale data without live integrations.	Data lead
E. Test and demo	Prove rule correctness, traceability, the block-to-release moment and recovery paths.	All
F. Jury presentation	Deliver a 20-minute consulting-grade story and answer senior questions.	Pitch lead

Playbook contents

Section	Topic
1	Executive proposition
2	Business problem and differentiation
3	Solution boundary and ownership
4	12-hour MVP scope
5	Reference enterprise scenario
6	Golden regulatory scenario
7	End-to-end compliance digital thread
8	Product Compliance Snapshot
9	Regulatory Change Implementation Register
10	Conformity, certification and evidence
11	Order-to-Delivery governance
12	Aftermarket extension
13	Enterprise orchestration
14	Lovable and Supabase architecture
15	Data model and server functions
16	Application screens and UX
17	12-hour execution plan
18	Synthetic dataset pack
19	Test cases and expected results
20	Acceptance criteria

Section	Topic
21	Demo reliability and recovery
22	20-minute jury script
23	Value case and responsible AI
24	Likely jury questions
25	Build checklists
Appendix A	Seed data tables
Appendix B	Detailed test catalogue
Appendix C	API and rule contracts
Appendix D	Lovable master build prompt

1. Executive proposition

The problem in one sentence

Regulatory compliance is not a document-reading problem; it is an enterprise execution and evidence problem.

Build an Enterprise Compliance Control Plane that converts a new or changed regulation into approved obligations, affected product scope, verified product/process/system changes, conformity and certification readiness, and an operational block-or-release decision.

What the MVP must prove

1. The platform owns regulatory content, versions, interpretations and obligations.
2. It identifies affected business units, product variants, markets, plants and configurations.
3. It creates and tracks product, process, system, governance, certification and aftermarket change requirements.
4. It does not treat a Jira task marked Done as proof of compliance; completion requires evidence and Compliance verification.
5. It owns conformity routes, certification programmes, certificates, evidence and certificate coverage.
6. It calculates a deterministic Product Compliance Snapshot that turns green only when every mandatory gate passes.
7. It blocks an affected order while mandatory certification or implementation evidence is missing, then releases the order after the blockers are resolved.
8. It shows the full digital thread from source clause to product, enterprise action, evidence, certificate, order and release token.

Jury closing message

We convert regulatory change into verified enterprise action and prevent a product from proceeding until the enterprise can prove that every mandatory product, process, system and certification change is complete.

2. Business problem and differentiation

A single regulatory amendment can drive changes across product requirements, BOM/configuration, calibration, suppliers, manufacturing, IT systems, certification, order controls, technical publications and the installed base. Existing tools often handle only one part of this chain.

Enterprise gap	Consequence	How the Control Plane responds
Regulatory interpretation is disconnected from delivery	Changes are missed or executed without traceability	Creates approved obligations and linked change requirements.
Applicability is assessed manually across BUs	Wrong products are included or affected variants are missed	Uses deterministic criteria plus human-reviewed AI recommendations.
Execution status lives in multiple tools	Leadership sees fragmented progress	Aggregates PLM, Jira, LeanIX, ServiceNow, ERP, MES and OM references.
Jira Done is treated as compliant	Implementation may lack evidence or verification	Requires evidence submission and Compliance verification.
Certificates are stored as documents only	Coverage by product, plant, calibration and date is unclear	Stores structured certificate master and coverage relationships.
Progress percentages hide critical blockers	A product can appear nearly complete but remain legally blocked	Uses mandatory gate logic; 96% complete can still be red.
Compliance status has no operational control	Orders or shipments may proceed despite gaps	Returns machine-readable BLOCK or RELEASE decisions.

Enterprise gap	Consequence	How the Control Plane responds
Aftermarket is assessed late	Installed products may remain exposed	Reuses applicability and evidence logic for installed assets.

What makes this different

- Not only regulatory summarisation: it creates obligations, applicability, change requirements and evidence needs.
- Not only a workflow engine: it has a compliance domain model covering conformity, certification and product-market authorisation.
- Not only a dashboard: snapshot status directly drives order and shipment decisions.
- Not a replacement for PLM, ERP, MES or OM: it provides the missing cross-enterprise compliance control plane.
- AI assists interpretation and recommendations; humans approve; deterministic rules decide readiness and release.

3. Solution boundary and ownership

3.1 Native capabilities owned by the platform

Native domain	Platform ownership
Regulatory Content	Regulations, amendments, versions, clauses, jurisdictions, dates and source files.
Regulatory Change Intelligence	Version comparison, added/modified/deleted clauses, interpretation and obligation extraction.
Product Applicability	BU, family, variant, market, plant, model year, configuration and exclusion decisions.
Compliance Requirements	Compliance-specific requirements and links to controlled PLM requirements.
Change Implementation	Product, process, system, documentation, governance, supplier, manufacturing, certification and aftermarket changes.
Conformity Management	Conformity routes, standards, assessment method and approval route.
Certification Management	Programmes, evidence checklists, submissions, authority review, issuance, renewals and status.
Certification Repository	Structured certificate master, document, coverage, validity, status, supersession and restrictions.
Evidence Management	Test reports, declarations, technical files, approvals, deployment proofs and production evidence.
Product Compliance Snapshot	Evidence-backed readiness for a specific product-market-plant-configuration context.
Compliance Policies	Deterministic readiness, market-authorisation and order-release rules.
Governance and Audit	Approvals, exceptions, waivers, reason codes, decisions, event history and traceability.
Aftermarket Compliance	Installed-base applicability and action assessment.

3.2 External enterprise systems

System	System-of-record responsibility
PLM / Requirements	Product definitions, BOMs, configurations, controlled product requirements and engineering changes.
ERP	Materials, suppliers, plants, inventory and commercial/production master data.
MES	Serialised manufacturing execution, tests and build evidence.
Order Management	Customer order, destination, requested configuration and delivery date.
Jira	Engineering, product and IT implementation work.
LeanIX	Capabilities, applications, interfaces, owners and lifecycle information.
ServiceNow	Operational remediation, incidents, exceptions and fulfilment tasks.
Legacy / Cloud apps	Test execution, supplier evidence, installed base, service execution and specialist product data.

Boundary rule

The Control Plane stores identifiers, compliance-relevant snapshots and verification evidence from enterprise systems; it does not duplicate complete BOMs, order masters or manufacturing histories.

4. Twelve-hour MVP scope

Dimension	Committed MVP scope
Regulation	One synthetic amendment represented by old and new versions.
Changed clauses	Five changes: certificate coverage, serial evidence, automated release control, field reassessment and retention.
Obligations	Three primary obligations plus one optional aftermarket obligation.
Business units	Engine, Components and Distribution in the golden path; Power Systems and Zero Emissions shown as excluded context.
Products	Ten families and twenty variants in seed data; one X15-D MY2027 US configuration in the golden path.
Markets	US, EU and India in data; the golden regulation applies primarily to US MY2027.
Sites	Multiple plants to demonstrate certificate coverage by manufacturing location.
Enterprise systems	PLM, ERP, MES, Order Management, Jira, LeanIX, ServiceNow and legacy/cloud applications - all mocked.
Certification	Two programmes, structured evidence checklists and multiple certificates with coverage differences.
Orders	Twenty-five synthetic orders with release, block, warning and not-applicable outcomes.
Installed base	Forty synthetic assets; one population assessment.
User experience	Six primary screens built in Lovable.
Backend	Supabase Postgres, Storage and server-side functions.

Dimension	Committed MVP scope
AI	Change explanation, obligation extraction, applicability suggestion and action recommendation.
Controls	Deterministic snapshot, certificate coverage and order gate logic.

4.1 Build deeply

- Regulatory content and version comparison.
- Obligation approval and applicability.
- Product Compliance Snapshot and change implementation tracking.
- Certification programme, evidence checklist and certificate coverage.
- Order block-to-release flow.
- Digital-thread traceability and one lightweight aftermarket assessment.

4.2 Simulate

- PLM requirement and engineering-change creation.
- Jira epic and task creation.
- LeanIX application-impact mapping.
- ServiceNow exceptions and remediation.
- ERP, MES and Order Management data exchange.
- Authority certificate issuance and supplier evidence retrieval.

4.3 Explicitly do not build

- Live customer integrations or external regulatory feeds.
- A generic legal interpretation engine for every jurisdiction.
- Full BOM recursion, workflow/BPM platform or complex identity administration.
- Authority submission integration, dealer campaign execution or multilingual NLP.
- Production-grade security and high-availability architecture.

5. Reference enterprise scenario

Use a synthetic global engine and power-product manufacturer. The enterprise context is intentionally broad enough to demonstrate cross-BU impact without claiming access to any customer-specific data.

BU	Hackathon role	Representative products
Engine	Primary product ownership and engineering	X15-D, X15-N, X10, B6.7
Components	Dependent product and certification scope	ATS-NG2, ECM-CM5, fuel system, turbo and sensors
Distribution	Technical publications, dealer/service and field actions	Service parts, bulletins, training and campaigns
Power Systems	Visible context and exclusion example	QSK60 power generation configuration
Zero Emissions	Visible context and exclusion example	BP-600 battery pack and EAX-14 eAxle

Assumed enterprise landscape

Capability	Representative application
Product lifecycle management	Global PLM
Requirements management	ReqHub
Enterprise resource planning	ERP-Core

Capability	Representative application
Manufacturing execution	Plant MES
Order management	Global OM
Supplier evidence	Supplier Connect
Testing and reports	TestVault
Installed base	Service Asset Hub
Delivery execution	Jira
Exceptions and incidents	ServiceNow
Architecture impact	LeanIX

6. Golden regulatory scenario

Synthetic regulation

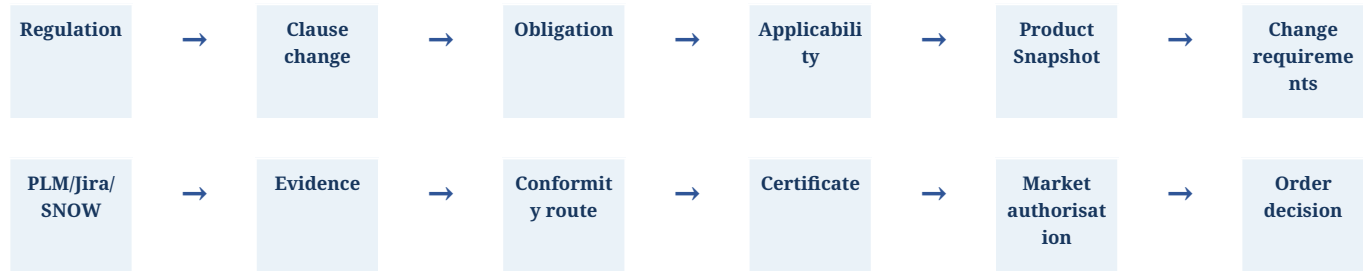
North American Heavy-Duty Emissions and Conformity Regulation 2027 - Version 2, effective 1 January 2027.

Obligation	Requirement	Key enterprise effect
OBL-001: Certificate coverage	An active certificate must cover engine family, aftertreatment, software calibration, manufacturing site, model year and market.	Structured certificate coverage and product-market authorisation.
OBL-002: Serial conformity evidence	Each shipped unit must have approved calibration, end-of-line evidence, serial configuration and site traceability.	MES evidence capture, retention and verification.
OBL-003: Automated release control	A product cannot be released or shipped until mandatory changes, certification, evidence and controls pass.	Order Management calls the Compliance Decision Service.
OBL-004: Installed-base reassessment	Material calibration or aftertreatment changes require an installed-base assessment.	Aftermarket population identification and action recommendation.

Example clause change

Clause	Old version	New version	Change
4.2	Certificate tied to the declared engine family.	Certificate must cover engine family, aftertreatment family, calibration range, plant and model year.	Modified
5.1	Periodic production evidence retained at programme level.	Serial-level production conformity evidence must be retained for each released unit.	Modified
6.4	No explicit automated release validation.	An automated compliance decision must occur before commercial release or shipment.	Added
7.3	Field reassessment may be performed when deemed necessary.	Field reassessment is mandatory after a material calibration or aftertreatment change.	Modified
9.2	Evidence retention: five years.	Evidence retention: ten years.	Modified

7. End-to-end Compliance Digital Thread



The Digital Thread is not merely a visual diagram. Each link is a stored relationship with an object ID, status, owner, source and audit history. A user must be able to click from an order blocker back to the certificate, evidence, change requirement, obligation and source clause.

Core thread relationships

From	Relationship	To
Regulation version	contains	Regulatory clause
Clause	creates / supports	Regulatory obligation
Obligation	applies to	Product variant
Obligation	drives	Change requirement
Change requirement	is implemented in	PLM/Jira/ServiceNow/legacy system
Change requirement	is verified by	Evidence artefact
Obligation	requires	Conformity route
Certification programme	collects	Evidence requirements
Certificate	covers	Product-market-plant-calibration scope
Product snapshot	governs	Order / shipment decision
Order decision	is explained by	Passed and failed control results
Installed asset	is assessed against	Obligation and configuration rules

8. Product Compliance Snapshot

Snapshot definition

A continuously calculated, evidence-backed compliance readiness view for Product + Variant + Market + Model Year + Plant + Configuration + Software/Calibration.

Regulatory	GREEN	All obligations interpreted, approved and applicable scope confirmed.
Product	GREEN	PLM requirements, configuration and calibration changes verified.
Process	AMBER	Order-to-Delivery process approval complete; training evidence pending.

Systems	RED	Order Management compliance gate not yet verified.
Certification	AMBER	Certificate in authority review.
Evidence	AMBER	MES deployment evidence submitted but not verified.
Operations	RED	Market authorisation inactive; affected orders must be blocked.

8.1 Readiness dimensions

Dimension	Mandatory checks
Regulatory	Applicable regulations identified; interpretation, obligation and applicability approved; unresolved questions closed.
Product	Requirements updated; engineering/BOM/configuration/calibration changes complete; validation approved.
Process	Process design, controls, work instructions, responsibilities, training and evidence capture operational.
Systems	Application/interface changes deployed, tested and Compliance-verified.
Certification	Conformity route approved; programme complete; submission resolved; certificate active and in scope.
Evidence	Every mandatory artefact exists, is approved, valid, correctly scoped and traceable.
Operations	Order/manufacturing/shipment controls active; no critical exception; product-market authorisation active.

8.2 Deterministic status logic

Status	Rule
GREEN	All applicable obligations approved; all mandatory changes Compliance Verified/Closed; all certificates and evidence valid; all controls active; no critical exception.
AMBER	Work is in progress before the deadline, verification is pending, or only non-blocking warnings/exceptions remain. Market release may still be unavailable.
RED	Any hard-stop condition exists: missing/expired/out-of-scope certificate, overdue blocking change, rejected evidence, inactive gate or critical exception.
GREY	Interpretation, applicability or impact assessment is incomplete.

Non-negotiable rule

Compliance is determined by mandatory gates, not average progress. A snapshot can be 96% complete and still be RED because one certificate or release control is missing.

9. Regulatory Change Implementation Register

Every required enterprise change is a first-class compliance object owned by the Control Plane. External systems execute work; the Control Plane owns the regulatory reason, completion criteria, evidence requirement and final verification.

9.1 Change categories

Category	Examples
Product	Requirement, design, BOM, component, material, calibration and label.
Process	Certification review, supplier approval, manufacturing check, order gate and evidence-retention process.
System	PLM attribute, ERP validation, MES evidence capture, OM rule, interface and reporting.
Certification	New certificate, extension, renewal, test programme and authority submission.
Documentation	Technical file, declaration, manual, installation instruction and service bulletin.
Governance	Approval gate, exception policy, control ownership and escalation.
Supplier / Manufacturing	Supplier declaration, PPAP, work instruction, tooling and end-of-line test.
Aftermarket	Population assessment, software update, campaign and part supersession.

9.2 Completion lifecycle



Additional states: Not Applicable, Blocked, Rejected, Exception Requested, Exception Approved and Overdue.

Critical distinction

A Jira task marked Done means delivery may be complete. It does not mean the regulatory change is compliant. Completion requires implementation evidence, evidence acceptance and Compliance verification.

9.3 Core change object

Change ID
 Source regulation / clause / obligation
 Affected Product Compliance Snapshot
 Category and required change
 Lifecycle stage / process / system
 Responsible function and owner
 Due date / criticality / blocking flag
 Execution system and reference
 Implementation status
 Evidence required and submitted
 Compliance verification
 Exception status and reason

10. Conformity, certification and evidence

Certification is native to the platform because it is the formal proof that approved obligations have been satisfied for a defined product scope. The module must be more than a document folder.

10.1 Certification lifecycle



10.2 Certification programme object

Programme ID
 Regulation and obligations
 Product-market-plant scope
 Conformity route / authority
 Owner and target date
 Evidence checklist
 Submission and review stage
 Certificate outcome
 Readiness impact

10.3 Structured certificate master

Certificate ID / number / type
 Issuing authority and regulatory scheme
 Products and variants covered
 Markets, plants and model years covered
 Software/calibration range
 Configuration restrictions
 Issue, effective and expiry dates
 Status and supersession
 Linked evidence and document

10.4 Certificate statuses

- Planned
- Evidence Pending
- Compliance Review
- Submitted
- Authority Review
- Issued
- Active
- Expiring
- Expired
- Suspended
- Withdrawn
- Superseded

10.5 Coverage evaluation

The order gate must not check only whether a certificate exists. It must verify that the certificate covers the exact product family, variant, market, model year, manufacturing plant, configuration, calibration and expected shipment date.

11. Order-to-Delivery governance

The operational proof point is an order that is blocked because the Product Compliance Snapshot or certificate coverage is incomplete, and is released after the missing requirements are verified.

11.1 Gate sequence

Gate	Checks	Example outcome
Order acceptance	Destination market, requested date, product family and regulation applicability.	Accept, conditional accept or hold.
Configuration	Approved product variant, BOM, software/calibration and certificate-scope attributes.	Configuration valid / invalid.
Supplier and material	Approved supplier, declarations, PPAP and controlled revisions.	Ready / evidence missing.
Manufacturing	Work instructions, controls, test readiness and serial evidence capture.	Build allowed / blocked.
Certification and release	Active certificate, technical file, mandatory evidence and no critical exception.	Market authorised / not authorised.
Shipment	Final snapshot, serial evidence and release token.	RELEASE or BLOCK.

11.2 Machine-readable decision

```
{
  "order_id": "ORD-001",
  "decision": "BLOCK",
  "reason_codes": ["CERTIFICATE_MISSING", "OM_GATE_NOT_VERIFIED"],
  "snapshot_id": "SNAP-PV-001",
  "obligation_ids": ["OBL-001", "OBL-003"],
  "remediation": ["Activate CERT-002", "Verify CHG-S-003"]
}
```

After evidence and change verification, the same order is re-evaluated and receives a release token such as CRT-ORD-001-2027.

12. Aftermarket extension

Aftermarket is deliberately thin in the 12-hour build. Reuse the same applicability and evidence logic against a synthetic installed-base dataset; do not build full campaign execution.



Possible outcome	MVP treatment
No impact	Record approved exclusion.
Documentation only	Create bulletin/document update action.
Calibration review	Create population and software assessment.
Dealer inspection	Create one campaign-assessment record.
Recall assessment	Show as future-state/high-risk workflow only.

13. Enterprise orchestration

System	What is triggered or read	What remains owned by the Control Plane
PLM	Create compliance requirement, impact record and engineering-change request.	Obligation link, compliance completion criteria and verification.
Jira	Create regulatory epic and engineering/IT tasks; read delivery status.	Change requirement and evidence acceptance.
LeanIX	Identify affected capabilities, apps, interfaces and owners.	Regulatory impact assessment and closure state.
ServiceNow	Create exceptions, evidence remediation and incidents.	Canonical compliance case, reason and final decision.
ERP	Read material, supplier, plant and status context.	Compliance interpretation and authorisation.
MES	Read serial test/configuration evidence.	Evidence requirement, approval and readiness impact.
Order Management	Read order/configuration; enforce hold or release.	Deterministic compliance decision and token.
Legacy / cloud apps	Read test, supplier, installed-base and service data.	Unified traceability and compliance status.

Adapter strategy

Build adapters, not integrations. For the hackathon, adapters read and write Supabase tables or JSON while presenting stable contracts that could later connect to real APIs.

14. Lovable and Supabase architecture

Lovable provides the unified compliance experience. Supabase provides the structured system of record, evidence storage and server-side decision services. GitHub sync should be enabled at project start so generated code can be reviewed and corrected.



14.1 Recommended stack

Layer	Technology	Purpose
Experience	Lovable / React	Enterprise dashboard, workflows and interactions.
Database	Supabase Postgres	Compliance domain objects and digital-thread relationships.
Document storage	Supabase Storage	Regulations, evidence, technical files and certificates.
Server logic	Supabase Edge Functions or server-side functions	LLM calls, rules, audit events and tokens.
AI	Approved enterprise LLM endpoint	Change interpretation, obligation extraction and recommendations.
Source control	GitHub sync	Code ownership, review, backup and manual fixes.
Mocks	Supabase tables / JSON	PLM, Jira, LeanIX, ServiceNow, ERP, MES and OM adapters.

14.2 Server-side only

- LLM keys and prompts.
- Readiness and certificate-coverage calculations.
- Order block/release rules and compliance-token generation.
- Evidence validity and audit-event creation.
- Approval authorization and state transitions.

14.3 Responsible AI split

Activity	Method
Explain clause changes	AI-assisted, source-grounded.
Extract proposed obligations	AI-assisted structured JSON.
Recommend applicability and actions	AI-assisted recommendation.
Approve interpretation/applicability	Human Compliance specialist.
Match exact product criteria	Deterministic rules.
Validate certificate coverage	Deterministic rules.
Calculate snapshot and order decision	Deterministic rules.
Override critical decision	Authorized human with recorded exception.

15. Data model and server functions

15.1 Minimum table groups

Domain	Tables
Regulatory	regulations, regulation_versions, regulatory_clauses, regulatory_changes, obligations, applicability_rules, applicability_decisions
Product	business_units, product_families, product_variants, plants, product_configurations, product_compliance_snapshots
Change	change_requirements, change_dependencies, execution_references, implementation_updates, verification_records, exceptions
Certification	conformity_routes, certification_programmes, evidence_requirements, evidence_artifacts, certification_submissions, certificates, certificate_coverage
Architecture and orchestration	applications, capabilities, integration_actions, jira_work_items, servicenow_cases, plm_requirements
Operations	orders, order_lines, compliance_decisions, compliance_tokens, manufacturing_records, installed_assets, aftermarket_assessments, audit_events

15.2 Functions to build

Function	Input	Output
analyse_regulatory_change	Old/new clause text	Change set and proposed obligations.
approve_obligation	Obligation + reviewer	Approved obligation and audit event.
assess_product_applicability	Obligation criteria + product variants	Applicable / excluded / conditional decisions with reasons.
generate_change_requirements	Approved obligation + impacted scope	Product/process/system/certification changes.
create_certification_programme	Obligations + product scope	Programme, route and evidence checklist.
evaluate_snapshot_readiness	Snapshot context and linked objects	Dimension statuses, overall status and blockers.
evaluate_certificate_coverage	Certificate + product/order context	Pass/fail with reason codes.
evaluate_order_compliance	Order + snapshot + evidence	BLOCK/RELEASE and token.
assess_aftermarket_population	Obligation + installed assets	Applicable, potential and excluded populations.

16. Application screens and UX

Screen	What the jury must see	Primary interaction
1. Executive Control Tower	Regulatory changes, impacted variants, critical changes, programmes at risk, orders on hold and deadlines.	Open the golden change.
2. Regulatory Content and Change	Old/new clause, highlighted change, AI interpretation, obligation and source traceability.	Approve OBL-001.
3. Product Impact and Snapshot	Affected/excluded products, readiness dimensions, blockers and change register.	Open X15-D US MY2027 snapshot.
4. Conformity and Certification	Route, programme, checklist, submission, certificate and structured coverage.	Approve final evidence / activate certificate.

Screen	What the jury must see	Primary interaction
5. Enterprise Orchestration	PLM requirement, Jira tasks, LeanIX impact and ServiceNow exception.	Verify system/process change.
6. Compliance Decision and Digital Thread	Order details, passed/failed controls, BLOCK/RELEASE, token and traceability.	Re-evaluate ORD-001; show thread and aftermarket.

16.1 Visual design

- Enterprise control-centre style: white background, dark blue navigation and restrained status colours.
- No oversized chatbot. AI interpretation appears as a contextual side panel with source clause and confidence.
- Every status has a reason; every blocker is clickable; every certificate shows coverage and validity.
- Use prominent BLOCKED and RELEASED states, readiness cards, evidence progress and a traceability timeline.
- Provide a Load Demo Scenario and Reset Scenario action.

17. Twelve-hour execution plan

Time	Workstream	Must-complete output
00:00-01:00	Freeze story and roles	Golden regulation, 3 obligations, 6 screens, demo order, data model and no-new-feature rule.
01:00-03:00	Shell, database and seed	Lovable navigation, Supabase schema, seed scripts, dashboard and status styling.
03:00-05:00	Regulatory and applicability	Clause comparison, obligation approval, deterministic product matching and exclusion reasons.
05:00-07:00	Snapshot and change register	Readiness dimensions, changes, status updates, evidence and recalculation.
07:00-09:00	Certification	Programme, route, checklist, certificate master, coverage and market authorisation.
09:00-10:30	Order gate and thread	Order selector, fail/pass controls, block, re-evaluate, release token and traceability.
10:30-11:00	Aftermarket and polish	Installed-base scan, one assessment and UX cleanup.
11:00-12:00	Freeze, test and rehearse	No new features; run acceptance suite; capture screenshots/video; rehearse 20-minute story.

17.1 Team roles

Role	Responsibilities
Product / pitch lead	Business logic, regulatory scenario, snapshot rules, presentation and demo narration.
Lovable frontend lead	UI, navigation, dashboard, interactions and visual consistency.
Supabase / rules lead	Schema, seed data, deterministic calculations, order gate and audit events.
AI / integration lead	Change analysis, structured responses, adapters, digital thread and automated tests.

18. Synthetic dataset pack

Seed more records than the golden path needs so the application feels enterprise-grade and supports negative, conditional and edge-case testing. The following tables are ready-to-use synthetic data definitions. Detailed rows are provided in Appendix A.

Dataset	Recommended count	Purpose
Business units	5	Cross-BU impact and exclusions.
Product families	10	Engine, component, power and zero-emission context.
Product variants	20	Market, model year, plant, configuration and calibration differences.
Regulatory clauses	5	Added and modified change detection.
Applicability rules	12	Applicable, excluded, conditional and aftermarket rules.
Change requirements	30	Product, process, system, certification, governance and aftermarket changes.
Evidence records	20	Approved, pending, rejected, missing and expired states.
Certificates	8	Active, pending, expired, suspended, wrong plant/calibration and future-effective coverage.
Applications	14	LeanIX-style impact landscape.
Jira work items	12	Done, blocked, in progress and done-but-unverified cases.
ServiceNow cases	8	Exceptions, evidence remediation and operational issues.
Orders	25	Block, release, warning, conditional and not-applicable outcomes.
Installed assets	40	Aftermarket population segmentation.
Test cases	72	Functional, negative, boundary, audit and recovery coverage.

19. Test strategy and expected results

Testing must prove deterministic compliance behavior, not only screen navigation. Use the golden path plus broad seed data to test negative and edge cases.

Test layer	What to prove
Regulatory intelligence	Added/modified/deleted clauses; source traceability; confidence and human approval.
Applicability	Correct inclusion/exclusion by market, year, fuel, plant, configuration and calibration.
Change tracking	Done is not verified; evidence rejection reopens work; blocking and non-blocking behavior.

Test layer	What to prove
Certification	Exact product/plant/calibration/date coverage; expiry, suspension, supersession and future-effective logic.
Snapshot	Mandatory gates override percentage; grey/amber/red/green calculations and blocker explanation.
Order gate	Release only after all mandatory controls pass; reason codes and audit events.
Aftermarket	Applicable population, potential manual assessment and exclusion.
Recovery	Cached AI response, reset scenario and deterministic seeded demo.

20. Non-negotiable acceptance criteria

#	Acceptance criterion
1	Two regulation versions and five changed clauses are visible.
2	At least one AI-proposed obligation can be human-approved.
3	Applicable, excluded and conditional products are calculated with reasons.
4	A Product Compliance Snapshot is created for the golden configuration.
5	Product, process, system and certification changes are visible and linked to obligations.
6	External execution references (PLM/Jira/LeanIX/ServiceNow) are visible.
7	Jira Done does not automatically complete a compliance change.
8	Evidence can be submitted, accepted/rejected and verified.
9	A certification programme, route and checklist are visible.
10	Certificate coverage includes product, plant, calibration, market and date.
11	Snapshot colours recalculate deterministically.
12	A non-compliant order is blocked with reason codes.
13	Completing the mandatory certificate/evidence and verifying the gate releases the order.
14	A compliance release token is generated only on pass.
15	The complete clause-to-order digital thread is clickable or visually traceable.
16	One aftermarket population assessment is shown.
17	The demo can be reset reliably and has a cached-AI fallback.
18	All critical actions create audit events.

21. Demo reliability and recovery

Mode	Use	Implementation
A. Live AI	Preferred demonstration of interpretation.	Call approved LLM endpoint and validate structured JSON.
B. Cached AI	Use when API latency/failure occurs.	Return previously validated JSON for the same clause set.
C. Fully seeded	Use for complete outage or timing risk.	Load demo scenario with all objects and known transitions.
Reset	Repeatable rehearsal and jury recovery.	Reset golden order, snapshot, certificate and change states.
Backup evidence	Fallback if UI fails.	Keep screenshots and a 2-minute screen recording available.

- Never rely on live Jira, LeanIX, ServiceNow or customer APIs.
- Use one known failing order and one known successful order.
- Freeze code at Hour 11; do not add features during rehearsal.
- Prepare presenter clicks and owner handoffs in advance.
- Validate the exact state transitions: Grey -> Red -> Green and BLOCK -> RELEASE.

22. Twenty-minute senior-jury script

Time	Presenter focus	What to say/show
0:00-2:00	Problem	"Regulatory compliance is not a document-reading problem; it is an enterprise execution problem." Show fragmentation across product, process, systems, certification and operations.
2:00-4:00	Solution	Introduce the Compliance Control Plane: native regulatory/certification system of record plus orchestration and deterministic control.
4:00-5:00	Architecture	Show one slide: Lovable experience, Supabase domain services, AI assistance, deterministic rules and enterprise adapters.
5:00-7:00	Regulatory demo	Open old/new clause, show change and approve OBL-001.
7:00-9:00	Impact demo	Show affected X15-D and ATS-NG2, excluded products and why.
9:00-11:00	Snapshot demo	Show RED/AMBER dimensions, blockers and change register. Highlight Jira Done but evidence unverified.
11:00-13:00	Certification demo	Show route, evidence checklist, pending certificate and structured coverage.
13:00-15:00	Operational demo	Evaluate ORD-001: BLOCK. Approve evidence/activate certificate/verify gate. Re-evaluate: RELEASE and token.
15:00-16:00	Digital thread	Navigate clause -> obligation -> product -> changes -> evidence -> certificate -> order.
16:00-17:00	Aftermarket	Show 40 assets scanned and the impacted population assessment.

Time	Presenter focus	What to say/show
17:00-18:00	Value + AI controls	Target outcomes and "AI recommends, humans approve, rules decide."
18:00-20:00	Scale and close	Explain adapter-based scaling and deliver closing line; allow brief questions if format requires.

Closing line

Regulus converts regulation into verified enterprise action and prevents a product from reaching the market until the enterprise can prove that every mandatory change is complete.

23. Value case and responsible AI

23.1 Value hypotheses

Value lever	Illustrative target hypothesis
Impact analysis	40-60% reduction in initial regulatory impact-analysis effort.
Coordination	30-50% reduction in compliance coordination and status-chasing effort.
Evidence readiness	Higher first-time completeness of certification and audit packages.
Operational control	Lower risk of a non-compliant order or shipment proceeding.
Auditability	Near-real-time traceability from source clause to product and evidence.
Reuse	Reusable obligation, applicability and certificate-coverage rules across products and markets.

Present these as hypotheses for future validation, not as proven benefits from the 12-hour prototype.

23.2 Responsible AI controls

Risk	Control
Hallucinated legal meaning	Show source clause, structured extraction, confidence and human approval.
Wrong applicability	Use explicit deterministic product criteria and approved exclusions.
AI release decision	Prohibited; release is rule-based.
Untraceable recommendation	Every AI output stores source, prompt version and reviewer.
Stale interpretation	Version regulations and record effective dates.
Unauthorized override	Require role-based approval and immutable audit reason.

Risk	Control
Sensitive enterprise data	Keep API keys server-side and use synthetic data in the hackathon.

24. Likely jury questions and crisp answers

Question	Answer
How is this different from regulatory-content tools?	Those tools identify and summarise change. This platform also owns conformity and certification, creates enterprise changes, verifies completion and controls market/order release.
Why own certification?	Certification is the formal proof that obligations are satisfied for a defined product, market, plant and configuration. Separating it breaks the digital thread.
Why not ServiceNow for everything?	ServiceNow can execute workflow and remediation, but the domain model linking regulation, product applicability, conformity, certificate coverage, evidence and market authorisation belongs in the Control Plane.
Why not PLM?	PLM owns product definition. Compliance crosses product, process, systems, certification, orders and installed assets.
Can AI release an order?	No. AI interprets and recommends; specialists approve; deterministic policy rules block or release.
How do you prevent hallucination?	Source grounding, structured outputs, confidence, human approval, deterministic criteria and complete audit history.
What is the innovation?	The evidence-backed Product Compliance Snapshot and its operational ability to block business execution until every mandatory change is verified.
How will it scale?	Through reusable compliance objects and adapter contracts connected to enterprise APIs without changing the core domain model.
What did you actually build in 12 hours?	A working golden path with native regulatory/certification records, synthetic enterprise adapters, deterministic readiness, order gating, digital thread and a seeded test suite.
What is not production-ready?	Real integrations, production security, broad legal content coverage, workflow hardening, performance and operating model.
Why Lovable?	It accelerates a polished enterprise experience while Supabase and reviewed server functions retain controlled logic and data ownership.
What happens when a Jira ticket is Done?	The change remains incomplete until required evidence is submitted, accepted and Compliance verifies that regulatory completion criteria are met.

25. Build and rehearsal checklists

25.1 Hour-1 scope freeze

Check	Done
Golden regulation and clauses frozen	☐
Three primary obligations frozen	☐
Golden product and order frozen	☐
Six screens frozen	☐
Snapshot gate rules frozen	☐
Team roles assigned	☐
No-new-feature rule accepted	☐
Fallback demo modes agreed	☐

25.2 Hour-11 technical freeze

Check	Done
All acceptance criteria tested	☐
Golden scenario reset works	☐
Cached AI response works	☐
Order block and release verified	☐
Audit records created	☐
Certificate coverage edge cases tested	☐
Screenshots captured	☐
Backup video captured	☐
Secrets server-side	☐
GitHub code committed	☐

25.3 Presenter rehearsal

Check	Done
20-minute run completed twice	☐

Check	Done
Presenter handoffs timed	☐
Demo clicks documented	☐
Jury Q&A owners assigned	☐
Architecture slide ready	☐
Value statements framed as hypotheses	☐
Synthetic-data disclaimer stated	☐
Closing line memorised	☐

Appendix A. Seed data tables

The following data can be seeded directly into Supabase or adapted into JSON/CSV. IDs are intentionally stable for demo scripts and test assertions.

A1. Business units

BU ID	Business unit	Role
BU-ENG	Engine	Engine design and manufacture
BU-CMP	Components	Aftertreatment, electronics and fuel systems
BU-DST	Distribution	Service, parts and field support
BU-PWR	Power Systems	Generator and high-horsepower systems
BU-ZE	Zero Emissions	Battery and electric powertrain

A2. Product families

Product ID	BU	Family	Fuel	Application
PF-001	BU-ENG	X15-D	Diesel	Heavy-duty on-highway
PF-002	BU-ENG	X15-N	Natural gas	Heavy-duty on-highway
PF-003	BU-ENG	X10	Diesel	Medium/heavy-duty
PF-004	BU-ENG	B6.7	Diesel	Medium-duty
PF-005	BU-CMP	ATS-NG2	N/A	Aftertreatment
PF-006	BU-CMP	ECM-CM5	N/A	Electronic control
PF-007	BU-CMP	FS-HPI	N/A	Fuel system
PF-008	BU-PWR	QSK60	Diesel	Generator/mining

Product ID	BU	Family	Fuel	Application
PF-009	BU-ZE	BP-600	Electric	Battery pack
PF-010	BU-ZE	EAX-14	Electric	eAxle

A3. Product variants

Variant	Family	Market	MY	Plant	Configuration	Software	Component
PV-001	PF-001	US	2027	Plant-A	CFG-X15-US-A	CAL-4.3	ATS-NG2
PV-002	PF-001	US	2027	Plant-B	CFG-X15-US-B	CAL-4.3	ATS-NG2
PV-003	PF-001	EU	2027	Plant-A	CFG-X15-EU-A	CAL-5.0	ATS-EU
PV-004	PF-001	India	2027	Plant-C	CFG-X15-IN-C	CAL-3.8	ATS-IN
PV-005	PF-002	US	2027	Plant-A	CFG-X15N-US	CAL-N2.1	ATS-NG
PV-006	PF-003	US	2027	Plant-B	CFG-X10-US	CAL-2.4	ATS-NG2
PV-007	PF-004	US	2026	Plant-B	CFG-B67-US	CAL-1.9	ATS-LG
PV-008	PF-005	US	2027	Plant-D	CFG-ATS-US	N/A	ATS-NG2
PV-009	PF-006	Global	2027	Plant-E	CFG-ECM-5	FW-8.2	N/A
PV-010	PF-008	US	2027	Plant-F	CFG-QSK-PWR	CAL-Q6	N/A
PV-011	PF-009	EU	2027	Plant-G	CFG-BAT-EU	SW-B4	N/A
PV-012	PF-010	US	2027	Plant-G	CFG-EAX-US	SW-E3	N/A
PV-013	PF-001	US	2028	Plant-A	CFG-X15-US-A2	CAL-4.5	ATS-NG2
PV-014	PF-001	Canada	2027	Plant-A	CFG-X15-CA-A	CAL-4.3	ATS-NG2
PV-015	PF-001	Mexico	2027	Plant-B	CFG-X15-MX-B	CAL-4.3	ATS-NG2
PV-016	PF-003	EU	2027	Plant-B	CFG-X10-EU	CAL-2.8	ATS-EU
PV-017	PF-002	US	2028	Plant-A	CFG-X15N-US2	CAL-N2.3	ATS-NG
PV-018	PF-004	US	2027	Plant-C	CFG-B67-US27	CAL-2.0	ATS-LG
PV-019	PF-005	EU	2027	Plant-D	CFG-ATS-EU	N/A	ATS-EU
PV-020	PF-006	US	2027	Plant-E	CFG-ECM-US	FW-8.3	N/A

A4. Applicability rules

Rule	Market	Product	MY	Fuel	Plant	Result	Reason/scope
AR-001	US	PF-001	2027	Diesel	Any	Applicable	OBL-001/2/3
AR-002	US	PF-005	2027	N/A	Any	Applicable	OBL-001/2
AR-003	EU	Any	Any	Any	Any	Not applicable	Jurisdiction exclusion

Rule	Market	Product	MY	Fuel	Plant	Result	Reason/scope
AR-004	US	PF-002	2027	Natural gas	Any	Conditional	Legal review required
AR-005	US	PF-008	Any	Diesel	Any	Not applicable	Power-generation exclusion
AR-006	US	PF-001	2026	Diesel	Any	Not applicable	Model year before 2027
AR-007	US	PF-003	2027	Diesel	Any	Applicable	Medium/heavy-duty scope
AR-008	US	PF-004	2027	Diesel	Any	Conditional	Weight-class assessment
AR-009	US	PF-001	2027	Diesel	Plant-A	Applicable	Plant-specific certificate required
AR-010	US	PF-001	2027	Diesel	Plant-B	Applicable	Separate plant coverage required
AR-011	US	Installed PF-001	2027	Diesel	Any	Aftermarket	OBL-004 assessment
AR-012	India	Any	Any	Any	Any	Not applicable	Jurisdiction exclusion

A5. Change requirements

Change	Category	Description	System	Owner	Block	Status	Evidence	Verification
CHG-P-001	Product	Add certificate coverage requirement to PLM	Global PLM	Product Compliance	Yes	Implemented	EV-001	Verified
CHG-P-002	Product	Update X15-D aftertreatment configuration	Global PLM	Product Engineering	Yes	Implemented	EV-002	Verified
CHG-P-003	Product	Approve calibration CAL-4.3	ReqHub/TestVault	Controls Engineering	Yes	Implemented	EV-003	Verified
CHG-P-004	Documentation	Update product label	Tech Docs	Technical Publications	No	In Progress	EV-016	Pending
CHG-P-005	Product	Update compliance specification	ReqHub	Systems Engineering	Yes	Implemented	EV-004	Verified
CHG-P-006	Manufacturing	Validate Plant-B configuration	Plant MES	Manufacturing Engineering	Yes	Blocked	EV-008	Not verified
CHG-PR-001	Process	Introduce order compliance review	Global OM	OM Process Owner	Yes	Implemented	EV-015	Verified
CHG-PR-002	Process	Add evidence-verification step	Compliance Platform	Compliance Operations	Yes	Implemented	EV-017	Verified
CHG-PR-003	Process	Update certification escalation process	Compliance Platform	Certification Lead	No	Planned	EV-018	Pending
CHG-PR-004	Supplier	Update supplier evidence workflow	Supplier Connect	Supplier Quality	Yes	Implemented	EV-005	Verified
CHG-PR-005	Process	Train plant release team	Learning Portal	Plant Quality	Yes	Evidence Submitted	EV-014	Pending
CHG-PR-006	Governance	Update exception approval matrix	Compliance Platform	Governance Lead	Yes	Implemented	EV-019	Verified

Change	Category	Description	System	Owner	Block	Status	Evidence	Verification
CHG-S-001	System	Add compliance requirement type	Global PLM	Engineering IT	Yes	Implemented	EV-001	Verified
CHG-S-002	System	Add certificate coverage API	Compliance Platform	Compliance IT	Yes	Implemented	EV-020	Verified
CHG-S-003	System	Call compliance gate before release	Global OM	Commercial IT	Yes	In Progress	EV-009	Rejected
CHG-S-004	System	Capture serial test evidence	Plant MES	Manufacturing IT	Yes	Implemented	EV-010	Pending
CHG-S-005	System	Add compliance status to ERP product view	ERP-Core	Enterprise Apps	No	Planned	None	Not verified
CHG-S-006	System	Add compliance epic template	Jira	Enterprise Tools	No	Implemented	EV-021	Verified
CHG-S-007	System	Add regulatory impact relation	LeanIX	Enterprise Architecture	No	Implemented	EV-022	Verified
CHG-S-008	System	Add compliance exception template	ServiceNow	Enterprise Tools	Yes	Implemented	EV-023	Verified
CHG-C-001	Certification	Create certification programme	Compliance Platform	Certification Lead	Yes	Closed	EV-024	Verified
CHG-C-002	Certification	Complete emissions testing	TestVault	Test Engineering	Yes	Closed	EV-004	Verified
CHG-C-003	Certification	Approve technical file	Compliance Platform	Compliance Reviewer	Yes	Closed	EV-011	Verified
CHG-C-004	Certification	Submit authority package	Compliance Platform	Certification Lead	Yes	Closed	EV-012	Verified
CHG-C-005	Certification	Issue and activate certificate	Compliance Platform	Certification Lead	Yes	Authority Review	EV-013	Not verified
CHG-G-001	Governance	Define market-authorisation control owner	Compliance Platform	Governance Lead	Yes	Closed	EV-025	Verified
CHG-G-002	Governance	Define retention policy for 10 years	Document Vault	Records Management	Yes	Implemented	EV-026	Pending
CHG-G-003	Governance	Define critical exception SLA	ServiceNow	Governance Lead	No	Closed	EV-027	Verified
CHG-A-001	Aftermarket	Assess installed X15-D CAL-4.3 population	Service Asset Hub	Aftermarket Compliance	Yes	Planned	None	Not verified
CHG-A-002	Aftermarket	Create dealer inspection decision	ServiceNow	Distribution BU	No	Identified	None	Not verified

A6. Evidence records

Evidence	Type	Status	Validity	Block	Scope	Obligation
EV-001	PLM requirement approval	Approved	Current	Yes	PV-001	OBL-001
EV-002	Engineering change	Approved	Current	Yes	PV-001	OBL-001

Evidence	Type	Status	Validity	Block	Scope	Obligation
EV-003	Calibration validation	Approved	Current	Yes	PV-001	OBL-001
EV-004	Emissions test report	Approved	Current	Yes	PV-001	OBL-001
EV-005	Supplier declaration	Approved	2028-12-31	Yes	ATS-NG2	OBL-001
EV-006	PPAP	Approved	2028-12-31	Yes	ATS-NG2	OBL-001
EV-007	Plant-A validation	Approved	Current	Yes	Plant-A	OBL-002
EV-008	Plant-B validation	Pending	N/A	Yes	Plant-B	OBL-002
EV-009	Order-gate test result	Rejected	N/A	Yes	Global OM	OBL-003
EV-010	MES interface deployment	Submitted	N/A	Yes	Plant MES	OBL-002
EV-011	Technical file	Approved	Current	Yes	PV-001	OBL-001
EV-012	Authority submission	Approved	Current	Yes	CERT-PROG-001	OBL-001
EV-013	Issued certificate	Missing	N/A	Yes	PV-001	OBL-001
EV-014	Training completion	Approved	2027-12-31	Yes	Plant-A	OBL-003
EV-015	Process approval	Approved	Current	Yes	Order-to-Delivery	OBL-003
EV-016	Product label proof	Pending	N/A	No	PV-001	OBL-001
EV-017	Evidence verification SOP	Approved	Current	Yes	Compliance Ops	OBL-002
EV-018	Escalation workflow	Missing	N/A	No	Certification	OBL-001
EV-019	Exception matrix	Approved	Current	Yes	Governance	OBL-003
EV-020	Coverage API test	Approved	Current	Yes	Compliance API	OBL-001

A7. Certificates

Certificate	Scope	Market	Plant	Calibration	Effective	Expiry	Status	Test purpose
CERT-001	X15-D Legacy US	US	Plant-A	CAL-4.2	2026-01-01	2028-12-31	Active	Does not cover CAL-4.3
CERT-002	X15-D MY27 US	US	Plant-A	CAL-4.3	2027-01-01	2029-12-31	Authority Review	Golden pending certificate
CERT-003	X15-D EU	EU	Plant-A	CAL-5.0	2026-01-01	2029-12-31	Active	EU scope
CERT-004	X10 MY27 US	US	Plant-B	CAL-2.4	2027-01-01	2029-12-31	Active	Exact coverage
CERT-005	X15-N US	US	Plant-A	CAL-N2.1	2025-01-01	2027-02-28	Expiring	Warning near expiry
CERT-006	B6.7 US	US	Plant-B	CAL-1.9	2025-01-01	2028-12-31	Active	MY2026 product
CERT-007	X15-D Plant-B	US	Plant-B	CAL-4.3	2027-06-01	2029-12-31	Issued	Future-effective for early orders
CERT-008	X15-D Suspended	US	Plant-A	CAL-4.3	2027-01-01	2029-12-31	Suspended	Cannot authorize release

A8. Applications

App ID	Application	Capability	Owner	Regulatory impact
APP-001	Global PLM	Product Lifecycle Management	Engineering IT	High
APP-002	ReqHub	Requirements Management	Engineering IT	High
APP-003	ERP-Core	Material/Plant Master	Enterprise Apps	Medium
APP-004	Plant MES	Manufacturing Execution	Manufacturing IT	Critical
APP-005	Global OM	Order Management	Commercial IT	Critical
APP-006	Supplier Connect	Supplier Evidence	Procurement IT	High
APP-007	Service Asset Hub	Installed Base	Service IT	Medium
APP-008	Jira	Delivery Execution	Enterprise Tools	Medium
APP-009	ServiceNow	Exceptions/Incidents	Enterprise Tools	High
APP-010	LeanIX	Architecture Impact	Enterprise Architecture	Medium
APP-011	TestVault	Validation Evidence	Engineering Tools	High
APP-012	Tech Docs	Product Documentation	Product Support IT	Medium
APP-013	Learning Portal	Training Evidence	HR IT	Low
APP-014	Document Vault	Records Retention	Compliance IT	High

A9. Jira work items

Jira	Change	Summary	Jira status	Evidence	Compliance verification
CMP-101	CHG-S-001	PLM compliance requirement type	Done	EV-001	Verified
CMP-102	CHG-S-003	Order compliance API and release call	In Progress	EV-009	Rejected
CMP-103	CHG-S-004	MES serial evidence interface	Done	EV-010	Pending
CMP-104	CHG-P-003	Calibration validation	Done	EV-003	Verified
CMP-105	CHG-P-006	Plant-B validation	Blocked	EV-008	Not verified
CMP-106	CHG-P-004	Product label update	In Progress	EV-016	Pending
CMP-107	CHG-PR-004	Supplier evidence workflow	Done	EV-005	Verified
CMP-108	CHG-S-008	ServiceNow exception template	Done	EV-023	Verified
CMP-109	CHG-G-002	10-year retention change	Done	EV-026	Pending
CMP-110	CHG-A-001	Installed-base assessment	To Do	None	Not verified
CMP-111	CHG-PR-005	Plant team training	Done	EV-014	Pending
CMP-112	CHG-C-005	Certificate activation workflow	In Progress	EV-013	Not verified

A10. ServiceNow cases

Case	Object	Summary	Status	Priority	Reason
SNOW-001	ORD-001	Order hold remediation	Open	Critical	CERTIFICATE_MISSING
SNOW-002	EV-009	Rejected order-gate evidence	Assigned	Critical	TEST_FAILED
SNOW-003	EV-010	MES evidence verification	Pending Review	High	EVIDENCE_UNVERIFIED
SNOW-004	PV-002	Plant-B certificate gap	Open	Critical	PLANT_NOT_COVERED
SNOW-005	CERT-005	Expiring certificate warning	Monitoring	Medium	CERT_EXPIRING
SNOW-006	CHG-G-002	Retention evidence review	Assigned	Medium	POLICY_EVIDENCE_PENDING
SNOW-007	ASSET-0007	Aftermarket manual assessment	New	High	CONFIG_UNKNOWN
SNOW-008	ORD-018	Approved non-blocking exception	Approved	Low	DOCUMENTATION_PENDING

A11. Orders

Order	Variant	Market	Plant	Software	Ship date	Expected	Reason
ORD-001	PV-001	US	Plant-A	CAL-4.3	2027-02-10	BLOCK	CERTIFICATE_MISSING; OM_GATE_NOT_VERIFIED
ORD-002	PV-001	US	Plant-A	CAL-4.2	2027-02-10	RELEASE	Covered by CERT-001; legacy config
ORD-003	PV-002	US	Plant-B	CAL-4.3	2027-03-01	BLOCK	PLANT_NOT_COVERED; PLANT_VALIDATION_PENDING
ORD-004	PV-003	EU	Plant-A	CAL-5.0	2027-02-01	RELEASE	EU certificate active
ORD-005	PV-006	US	Plant-B	CAL-2.4	2027-02-10	RELEASE	CERT-004 exact coverage
ORD-006	PV-007	US	Plant-B	CAL-1.9	2026-11-01	NOT_APPLICABLE	Model year before 2027
ORD-007	PV-010	US	Plant-F	CAL-Q6	2027-02-01	NOT_APPLICABLE	Power Systems excluded
ORD-008	PV-005	US	Plant-A	CAL-N2.1	2027-02-15	WARNING	Conditional applicability; certificate expiring
ORD-009	PV-001	US	Plant-A	CAL-4.3	2027-04-10	BLOCK	SERIAL_EVIDENCE_MISSING
ORD-010	PV-001	US	Plant-A	CAL-4.3	2027-04-10	RELEASE_AFTER_FIX	Golden post-activation outcome
ORD-011	PV-013	US	Plant-A	CAL-4.5	2028-02-10	BLOCK	CALIBRATION_NOT_COVERED
ORD-012	PV-014	Canada	Plant-A	CAL-4.3	2027-05-10	CONDITIONAL	Jurisdiction mapping review
ORD-013	PV-015	Mexico	Plant-B	CAL-4.3	2027-05-10	CONDITIONAL	Jurisdiction mapping review
ORD-014	PV-018	US	Plant-C	CAL-2.0	2027-03-20	CONDITIONAL	Weight-class applicability
ORD-015	PV-011	EU	Plant-G	SW-B4	2027-03-20	NOT_APPLICABLE	Zero Emissions product
ORD-016	PV-012	US	Plant-G	SW-E3	2027-03-20	NOT_APPLICABLE	eAxle excluded

Order	Variant	Market	Plant	Software	Ship date	Expected	Reason
ORD-017	PV-001	US	Plant-A	CAL-4.3	2026-12-15	BLOCK	Certificate not effective on ship date
ORD-018	PV-001	US	Plant-A	CAL-4.3	2027-06-15	CONDITIONAL_RELEASE	Non-blocking documentation exception approved
ORD-019	PV-001	US	Plant-A	CAL-4.3	2027-06-15	BLOCK	Critical exception unresolved
ORD-020	PV-002	US	Plant-B	CAL-4.3	2027-07-15	RELEASE	CERT-007 effective and plant covered
ORD-021	PV-001	US	Plant-A	CAL-4.3	2029-01-10	BLOCK	CERT-002 expired
ORD-022	PV-001	US	Plant-A	CAL-4.3	2027-03-10	BLOCK	CERT-008 suspended
ORD-023	PV-006	US	Plant-B	CAL-2.4	2027-03-10	BLOCK	Mandatory evidence rejected
ORD-024	PV-001	US	Plant-A	CAL-4.3	2027-03-10	BLOCK	Snapshot still grey - applicability review incomplete
ORD-025	PV-003	India	Plant-C	CAL-3.8	2027-03-10	NOT_APPLICABLE	Jurisdiction exclusion

A12. Installed assets

Asset	Family	MY	Market	Plant	Calibration	Aftertreatment	Assessment
ASSET-0001	PF-004	2026	US	Plant-B	CAL-1.8	ATS-LG	Not applicable
ASSET-0002	PF-001	2025	US	Plant-C	CAL-4.3	ATS-NG2	Not applicable
ASSET-0003	PF-001	2026	US	Plant-A	CAL-4.2	ATS-NG2	Not applicable
ASSET-0004	PF-001	2027	Mexico	Plant-C	CAL-4.2	UNKNOWN	Manual assessment
ASSET-0005	PF-002	2026	US	Plant-C	CAL-N2.1	ATS-NG	Not applicable
ASSET-0006	PF-001	2025	Canada	Plant-B	CAL-4.2	UNKNOWN	Not applicable
ASSET-0007	PF-001	2027	US	Plant-C	UNKNOWN	ATS-NG1	Manual assessment
ASSET-0008	PF-001	2026	US	Plant-C	CAL-4.2	UNKNOWN	Not applicable
ASSET-0009	PF-003	2025	US	Plant-B	CAL-2.3	ATS-NG2	Not applicable
ASSET-0010	PF-001	2027	US	Plant-A	UNKNOWN	ATS-NG1	Manual assessment
ASSET-0011	PF-002	2025	US	Plant-C	CAL-N2.1	ATS-NG	Not applicable
ASSET-0012	PF-001	2026	Canada	Plant-B	CAL-4.3	ATS-NG1	Not applicable
ASSET-0013	PF-001	2025	US	Plant-A	CAL-4.3	ATS-NG2	Not applicable
ASSET-0014	PF-003	2025	Mexico	Plant-A	CAL-2.3	ATS-NG2	Not applicable
ASSET-0015	PF-001	2027	Canada	Plant-C	CAL-4.2	ATS-NG2	Not applicable
ASSET-0016	PF-001	2025	Canada	Plant-B	UNKNOWN	ATS-NG1	Not applicable

Asset	Family	MY	Market	Plant	Calibration	Aftertreatment	Assessment
ASSET-0017	PF-001	2026	Mexico	Plant-C	UNKNOWN	ATS-NG1	Not applicable
ASSET-0018	PF-001	2026	Canada	Plant-B	UNKNOWN	ATS-NG1	Not applicable
ASSET-0019	PF-003	2025	US	Plant-A	CAL-2.4	ATS-NG2	Not applicable
ASSET-0020	PF-003	2027	US	Plant-A	CAL-2.3	ATS-NG2	Not applicable
ASSET-0021	PF-001	2025	US	Plant-C	UNKNOWN	ATS-NG1	Not applicable
ASSET-0022	PF-001	2027	Canada	Plant-B	UNKNOWN	ATS-NG2	Manual assessment
ASSET-0023	PF-004	2025	US	Plant-B	CAL-1.8	ATS-LG	Not applicable
ASSET-0024	PF-003	2025	US	Plant-B	CAL-2.3	ATS-NG2	Not applicable
ASSET-0025	PF-004	2026	Canada	Plant-A	CAL-1.9	ATS-LG	Not applicable
ASSET-0026	PF-001	2025	US	Plant-A	CAL-4.3	ATS-NG1	Not applicable
ASSET-0027	PF-001	2026	Mexico	Plant-B	CAL-4.2	ATS-NG1	Not applicable
ASSET-0028	PF-003	2026	US	Plant-A	CAL-2.4	ATS-NG2	Not applicable
ASSET-0029	PF-001	2026	Mexico	Plant-B	CAL-4.2	ATS-NG2	Not applicable
ASSET-0030	PF-001	2027	US	Plant-C	CAL-4.2	UNKNOWN	Manual assessment
ASSET-0031	PF-004	2026	Mexico	Plant-C	CAL-1.8	ATS-LG	Not applicable
ASSET-0032	PF-001	2025	Canada	Plant-C	UNKNOWN	UNKNOWN	Not applicable
ASSET-0033	PF-004	2025	US	Plant-B	CAL-1.9	ATS-LG	Not applicable
ASSET-0034	PF-001	2025	US	Plant-A	CAL-4.3	ATS-NG1	Not applicable
ASSET-0035	PF-001	2025	US	Plant-C	UNKNOWN	UNKNOWN	Not applicable
ASSET-0036	PF-001	2025	Canada	Plant-A	UNKNOWN	UNKNOWN	Not applicable
ASSET-0037	PF-003	2026	Mexico	Plant-A	CAL-2.3	ATS-NG2	Not applicable
ASSET-0038	PF-001	2027	US	Plant-A	UNKNOWN	UNKNOWN	Manual assessment
ASSET-0039	PF-003	2026	US	Plant-B	CAL-2.3	ATS-NG2	Not applicable
ASSET-0040	PF-002	2027	US	Plant-A	CAL-N2.0	ATS-NG	Not applicable

Appendix B. Detailed test catalogue

#	Category	Test	Input / setup	Expected result
1	Regulatory	New clause detection	Clause 6.4 present only in V2	Added change set created with source.
2	Regulatory	Modified clause detection	Clause 4.2 wording differs	Modified change set created.
3	Regulatory	Deleted clause detection	Seed a V1-only clause	Deleted change set created.
4	Regulatory	Source traceability	Approve OBL-001	Obligation retains regulation/version/clause IDs.

#	Category	Test	Input / setup	Expected result
5	Regulatory	Low confidence	Return confidence < threshold	Human review required; no downstream creation.
6	Regulatory	Rejected interpretation	Reviewer rejects proposed obligation	No applicability or changes generated.
7	Regulatory	Approved interpretation	Reviewer approves OBL-001	Applicability becomes available.
8	Regulatory	Version effective date	Compare order date vs regulation date	Correct version applied.
9	Applicability	US MY2027 diesel	PV-001	Applicable.
10	Applicability	EU variant	PV-003	Excluded by jurisdiction.
11	Applicability	MY2026 product	PV-007	Excluded by model year.
12	Applicability	Dependent aftertreatment	PV-008	Applicable through component dependency.
13	Applicability	Power Systems	PV-010	Excluded.
14	Applicability	Natural-gas variant	PV-005	Conditional.
15	Applicability	Plant-A	PV-001	Applicable; plant coverage required.
16	Applicability	Plant-B	PV-002	Applicable; separate plant certificate required.
17	Applicability	Unknown configuration	Asset with UNKNOWN	Manual assessment.
18	Applicability	Software version exact	CAL-4.3	Applicable.
19	Applicability	Wrong calibration	CAL-4.2 under new scope	Not covered by CERT-002.
20	Applicability	India market	PV-004	Excluded by jurisdiction.
21	Change	Jira Done without evidence	CMP-103	Change remains incomplete.
22	Change	Evidence submitted	EV-010 Submitted	Still incomplete until verified.
23	Change	Verified evidence	Verify EV-010	Change completes.
24	Change	Approved N/A	Approve exclusion	Counts as closed.
25	Change	Unapproved N/A	Set N/A without approval	Remains open.
26	Change	Overdue blocker	CHG-S-003 past due	Snapshot RED.
27	Change	Non-blocking open	CHG-P-004 open	Snapshot AMBER if no hard stop.
28	Change	Critical exception	Open critical exception	Snapshot RED.
29	Change	Rejected evidence	EV-009 Rejected	Change reopens / blocks.
30	Change	Status recalculation	Verify final blocker	Snapshot recalculates immediately.
31	Change	Dependency	Certificate change depends on test	Certificate cannot close first.
32	Change	Owner escalation	Overdue critical change	ServiceNow escalation created.
33	Certification	Exact active coverage	CERT-004 + ORD-005	Pass.
34	Certification	Wrong calibration	CERT-001 + CAL-4.3	Fail.
35	Certification	Wrong plant	CERT-002 + Plant-B	Fail.
36	Certification	Expired certificate	CERT-002 after expiry	Fail.
37	Certification	Suspended certificate	CERT-008	Fail.
38	Certification	Expiring certificate	CERT-005	Warning.

#	Category	Test	Input / setup	Expected result
39	Certification	Missing technical file	EV-011 missing	Programme not issuable.
40	Certification	Active certificate	Activate CERT-002	Market authorisation updates.
41	Certification	Superseded certificate	Mark certificate superseded	Fail.
42	Certification	Future effective	CERT-007 before 2027-06-01	Fail.
43	Certification	Configuration restriction	Unsupported aftertreatment	Fail.
44	Certification	Market mismatch	EU cert used for US order	Fail.
45	Order gate	Compliant order	ORD-005	RELEASE.
46	Order gate	Missing certificate	ORD-001 initial	BLOCK.
47	Order gate	Missing evidence	ORD-009	BLOCK.
48	Order gate	Missing system gate	CHG-S-003 not verified	BLOCK.
49	Order gate	Not applicable	ORD-006	Proceed as NOT_APPLICABLE.
50	Order gate	Plant not covered	ORD-003	BLOCK.
51	Order gate	Critical exception	ORD-019	BLOCK.
52	Order gate	Non-blocking exception	ORD-018	Conditional release.
53	Order gate	Re-evaluation	Fix ORD-001 blockers	RELEASE.
54	Order gate	Token generation	All checks pass	Token issued once.
55	Order gate	Reason codes	Any block	All failed controls stored.
56	Order gate	Idempotency	Re-run released order	Same valid decision/token semantics.
57	Order gate	Effective-date check	ORD-017	BLOCK.
58	Order gate	Suspension event	Suspend cert after release	Future orders block; audit event created.
59	Snapshot	All green	All mandatory items pass	Overall GREEN.
60	Snapshot	One red dimension	Certificate missing	Overall RED.
61	Snapshot	96% complete	Only mandatory cert missing	Overall RED.
62	Snapshot	Assessment incomplete	Applicability pending	GREY.
63	Snapshot	In progress	No hard stop before deadline	AMBER.
64	Snapshot	Blocker explanation	Open snapshot	Exact blockers and links visible.
65	Snapshot	Traceability	Click blocker	Navigates to obligation/evidence.
66	Snapshot	Immediate refresh	Verify change	Status recalculates.
67	Snapshot	Non-blocking warning	Label update pending	AMBER or GREEN per policy.
68	Snapshot	Approved critical waiver	Policy disallows green	Status remains AMBER/RED as defined.
69	Aftermarket	Applicable asset	PF-001/MY2027/US/CAL-4.3/ATS-NG2	Applicable.
70	Aftermarket	Wrong year	MY2026	Excluded.
71	Aftermarket	Wrong calibration	CAL-4.2	Excluded or separate review.
72	Aftermarket	Unknown configuration	UNKNOWN	Manual assessment.

#	Category	Test	Input / setup	Expected result
73	Aftermarket	Assessment record	Applicable population	ServiceNow assessment created.
74	Aftermarket	Completed action no evidence	Service done, evidence missing	Asset status AMBER.
75	Aftermarket	Country conditional	Canada/Mexico	Conditional review.
76	Aftermarket	Duplicate asset event	Same event reprocessed	No duplicate campaign assessment.
77	Recovery / audit	AI timeout	Live endpoint unavailable	Cached JSON used.
78	Recovery / audit	Database reset	Click Reset Scenario	Golden initial state restored.
79	Recovery / audit	Repeated demo	Run block-to-release twice	Consistent outcome.
80	Recovery / audit	Missing document file	Metadata exists; binary missing	Evidence invalid and block shown.
81	Recovery / audit	Unauthorized approval	Non-compliance role	Approval denied and logged.
82	Recovery / audit	Audit integrity	Change state transition	Event records actor, time, before/after and reason.
83	Recovery / audit	Partial adapter failure	Jira mock fails	Compliance change remains open; retry visible.
84	Recovery / audit	Frontend refresh	Reload page	State remains persisted.

Appendix C. API and rule contracts

C1. Obligation extraction response

```
{
  "obligation_id": "OBL-001",
  "source_clause_id": "CLAUSE-4.2-V2",
  "requirement": "An active certificate shall cover engine, aftertreatment, calibration, plant and model year.",
  "jurisdiction": "US",
  "effective_date": "2027-01-01",
  "applicability": {
    "product_family": ["PF-001", "PF-003", "PF-005"],
    "model_year_min": 2027,
    "market": ["US"]
  },
  "required_evidence_types": ["technical_file", "test_report", "certificate"],
  "confidence": 0.91,
  "review_status": "PROPOSED"
}
```

C2. Snapshot evaluation pseudocode

```

if applicability_not_approved:
    overall = GREY
elif any(hard_stop_condition):
    overall = RED
elif all(mandatory_changes_verified) \
    and all(required_certificates_active_and_in_scope) \
    and all(mandatory_evidence_approved_and_valid) \
    and no_critical_exception \
    and operational_controls_active:
    overall = GREEN
else:
    overall = AMBER

```

C3. Certificate coverage function

```

evaluate_certificate_coverage(certificate, context):
    check status == ACTIVE
    check effective_date <= requested_date <= expiry_date
    check market in certificate.markets
    check product_variant / family coverage
    check manufacturing_plant coverage
    check model_year coverage
    check calibration range
    check aftertreatment / configuration restrictions
    return PASS or FAIL + reason_codes

```

C4. Order decision response

```

{
  "order_id": "ORD-001",
  "decision": "RELEASE",
  "snapshot_status": "GREEN",
  "passed_controls": ["APPLICABILITY_APPROVED", "CERTIFICATE_COVERAGE", "MANDATORY_EVIDENCE", "NO_CRITICAL_EXCEPTION", "OM_GATE_ACTIVE"],
  "failed_controls": [],
  "compliance_token": "CRT-ORD-001-2027",
  "decision_timestamp": "<server timestamp>",
  "audit_event_id": "AUD-..."
}

```

Appendix D. Lovable master build prompt

Use this as the starting prompt, then build and validate one module at a time. Keep GitHub sync enabled and review generated code. Do not ask Lovable to invent the business rules; provide the deterministic rules from this playbook.

Build an enterprise web application called "Regulus - Enterprise Compliance Control Plane" using React and Supabase. The app is a unified experience and compliance system of record for regulatory content, obligations, product applicability, change implementation, conformity, certification, evidence, Product Compliance Snapshots, order decisions and digital-thread traceability.

DESIGN

- Enterprise consulting-grade UI: white background, dark navy navigation, restrained blue/grey palette and clear red/amber/green status states.
- Left navigation with: Control Tower, Regulatory Change, Product Impact & Snapshot, Certification, Enterprise Orchestration, Compliance Decision & Digital Thread.
- Dense but readable tables, filter panels, status cards, audit metadata and clickable blockers.
- Do not create a large chatbot UI. AI appears only as a contextual interpretation panel with source clause and confidence.

DATA AND BACKEND

- Use Supabase Postgres and Storage. Create the table groups and seed data defined in the playbook.
- Keep AI calls, snapshot rules, certificate coverage rules, order release rules, audit events and token generation server-side.
- Add a Load Demo Scenario and Reset Demo Scenario action.
- Create mock adapters for PLM, Jira, LeanIX, ServiceNow, ERP, MES and Order Management using Supabase tables.

SCREEN BEHAVIOR

1. Control Tower: show counts for changed regulations, impacted variants, critical changes, certification programmes at risk, non-authorised products, orders on hold and upcoming deadlines.
2. Regulatory Change: show old/new clause, highlighted change, AI-proposed obligation, source and confidence. Allow Compliance approval.
3. Product Impact & Snapshot: show affected/excluded products with reasons. Show seven readiness dimensions, overall status, blockers and change register. A Jira Done item is not complete until evidence is accepted and Compliance Verified.
4. Certification: show conformity route, programme, evidence checklist, submission, certificate master and exact coverage by product, market, plant, model year, configuration, calibration and date.
5. Enterprise Orchestration: show PLM requirement, Jira work, LeanIX impacts and ServiceNow exceptions. Allow verification of evidence and changes.
6. Compliance Decision & Digital Thread: select an order, evaluate deterministic controls, show BLOCK or RELEASE, reason codes, token and the full clause-to-order thread. Include a small installed-base assessment panel.

GOLDEN DEMO

- ORD-001 is initially BLOCKED because CERT-002 is not active and CHG-S-003 is not Compliance Verified.
- CMP-103 is Jira Done but CHG-S-004 remains incomplete because EV-010 is only Submitted.
- When final evidence is approved, CERT-002 becomes Active and CHG-S-003/CHG-S-004 become Compliance Verified, the Product Compliance Snapshot recalculates GREEN.
- Re-evaluating ORD-001 returns RELEASE and generates CRT-ORD-001-2027.

RULES

- AI interprets and recommends. Human Compliance reviewers approve. Deterministic rules calculate applicability, certificate coverage, snapshot status and order decision.
- Mandatory gates override progress percentage. 96% complete with a missing mandatory certificate remains RED.
- Every state change creates an audit event with actor, timestamp, before/after state and reason.

Use realistic seeded data, clear empty/error states and a reliable reset flow.

Final team alignment

Build this - and nothing broader

One regulation changes. The platform understands the obligation, identifies the affected product, creates and verifies enterprise changes, owns conformity and certification, blocks a non-compliant order, then releases it after every mandatory gate is proven. The Digital Thread explains exactly why.

Success is not the number of screens or AI calls. Success is a controlled, repeatable block-to-release demonstration that senior jurors can immediately map to a real manufacturing enterprise.

END OF PLAYBOOK